

20. The CAPM [see Exercises 10.14 and 2.16] says that the risk premium on security j is related to the risk premium on the market portfolio. That is $r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$ where r_j and r_f are the returns to security j and the risk-free rate, respectively, r_m is the return on the market portfolio, and β_j is the j th security's "beta" value. We measure the market portfolio using the Standard & Poor's value weighted index, and the risk-free rate by the 30-day LIBOR monthly rate of return. As noted in Exercise 10.14, if the market return is measured with error, then we face an errors-in-variables, or measurement error, problem.

a. Use the observations on Microsoft in the data file *capm5* to estimate the CAPM model using OLS. How would you classify the Microsoft stock over this period? Risky or relatively safe, relative to the market portfolio?

We estimate the CAPM model for Microsoft using OLS:

$$r_j - r_f = \alpha + \beta(r_m - r_f) + u$$

For Microsoft, the estimated equation is:

$$msft - riskfree = 0.00325 + 1.20184(mkt - riskfree)$$

The estimated beta for Microsoft is 1.2018.

Since beta is greater than 1, Microsoft stock appears to be riskier than the market portfolio over this period. This means Microsoft's returns are more sensitive to changes in market returns. The beta coefficient is highly statistically significant ($t = 9.839$), indicating a strong positive relationship between Microsoft and market excess returns.

b. It has been suggested that it is possible to construct an IV by ranking the values of the explanatory variable and using the rank as the IV, that is, we sort $(r_m - r_f)$ from smallest to largest, and assign the values $RANK = 1, 2, \dots, 180$. Does this variable potentially satisfy the conditions IV1-IV3?

Create *Rank* and obtain the first-stage regression results. Is the coefficient of *RANK* very significant? What is the R^2 of the first-stage regression? Can *RANK* be regarded as a strong IV?

RANK is likely to satisfy IV1 because it is highly correlated with market excess returns. It may also satisfy IV2 if ranking the market excess return does not create correlation with the regression error term. IV3 is satisfied because *RANK* varies across observations and is not a constant.

We create *RANK* by sorting the values of $(r_m - r_f)$ from smallest to largest and assigning values from 1 to 180. The first-stage regression is:

$$mkt_excess = \pi_0 + \pi_1 RANK + v$$

The estimated first-stage equation is:

$$mkt_excess = -0.0790 + 0.0009067 RANK$$

The coefficient of *RANK* is highly statistically significant with a t-statistic of 43.1 and a p-value less than 0.001. The R^2 of the first-stage regression is 0.9126. The F-statistic is 1858, which is far greater than the rule-of-thumb value of 10.

Therefore, *RANK* can be regarded as a very strong instrumental variable (strong IV).

c. Compute the first-stage residuals, $\hat{v}$, and add them to the CAPM model. Estimate the resulting augmented equation by OLS and test the significance of $\hat{v}$ at 1% level of significance. Can we conclude that the market return is exogenous?

We first obtain the first-stage residuals $\hat{v}$ from:

$$mkt_excess = \pi_0 + \pi_1 RANK + v$$

Then we estimate the augmented CAPM equation:

$$msft_excess = \alpha + \beta mkt_excess + \delta \hat{v} + e$$

The estimated coefficient on $\hat{v}$ is -0.8746, with a t-statistic of -2.040 and a p-value of 0.0428. Since the p-value is greater than 0.01, we fail to reject the null hypothesis that $\delta = 0$ at the 1% significance level. Therefore, we cannot reject the exogeneity of the market return at the 1% level.

d. Use $RANK$ as an IV and estimate the CAPM model by IV/2SLS. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

We estimate the CAPM model using $RANK$ as an instrumental variable via 2SLS:

$$msft_excess = \alpha + \beta mkt_excess + u$$

The IV estimate of beta is 1.2783, while the OLS estimate from part (a) is 1.2018. Compared to OLS, the IV estimate is larger in magnitude. The IV estimate suggests that Microsoft is slightly more sensitive to market movements than indicated by OLS.

This result is consistent with the presence of measurement error in the market return. Measurement error in the explanatory variable typically leads to attenuation bias, causing the OLS estimate to be biased toward zero. The IV estimator corrects this bias and produces a larger estimate of beta.

e. Create a new variable $POS = 1$ if the market return $(r_m - r_f)$ is positive, and zero otherwise. Obtain the first-stage regression results using both $RANK$ and POS as instrumental variables. Test the joint significance of the IV. Can we conclude that we have adequately strong IV? What is the R^2 of the first-stage regression?

We estimate the first-stage regression:

$$mkt_excess = \pi_0 + \pi_1 RANK + \pi_2 POS + v$$

To test the joint significance of the instruments, we test:

$$H_0: \pi_1 = \pi_2 = 0$$

The F-statistic is 951.3 with a p-value less than 0.001. Therefore, we reject the null hypothesis that the instruments are jointly insignificant. The R^2 of the first-stage regression is 0.9149.

Since the F-statistic is far greater than the rule-of-thumb value of 10, we conclude that $RANK$ and POS are strong instrumental variables.

f. Carry out the Hausman test for endogeneity using the residuals from the first-stage equation in (e).

Can we conclude that the market return is exogenous at the 1% level of significance?

We perform a Hausman test by including the first-stage residuals from the regression of *mkt_excess* on *RANK* and *POS* into the CAPM equation:

$$msft_excess = \alpha + \beta mkt_excess + \delta \hat{v} + e$$

We test the null hypothesis:

$$H_0: \delta = 0$$

The coefficient on $\hat{v}$ is -0.9549 with a p-value of 0.0287. Since the p-value is greater than 0.01, we fail to reject the null hypothesis. Therefore, we conclude that there is no evidence of endogeneity in the market return at the 1% significance level.

g. Obtain the IV/2SLS estimates of the CAPM model using *RANK* and *POS* as instrumental variables. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

We estimate the CAPM model using *RANK* and *POS* as instrumental variables via 2SLS:

$$msft_excess = \alpha + \beta mkt_excess + u$$

The IV estimate of beta is 1.2831, while the OLS estimate from part (a) is 1.2018.

Compared to OLS, the IV estimate is larger in magnitude. This result is consistent with the presence of measurement error in the market return, which leads to attenuation bias in OLS estimates (bias toward zero). The IV estimator corrects this bias and produces a higher and more consistent estimate of beta.

The result is also consistent with expectations because the similarity between the IV estimates using *RANK* alone and *RANK* + *POS* suggests that *RANK* contains most of the explanatory power in the first stage.

h. Obtain the IV/2SLS residuals from part (g) and use them (not an automatic command) to carry out a Sargan test for the validity of the surplus IV at the 5% level of significance.

We perform a Sargan test for the validity of the surplus instrumental variables using the IV residuals from part (g). The test statistic is computed as:

$$S = n \cdot R^2 = 0.5600$$

Under the null hypothesis that the instruments are valid, the test statistic follows a chi-square distribution with 1 degree of freedom. The 5% critical value is 3.8414. Since the test statistic is smaller than the critical value, we fail to reject the null hypothesis.

Therefore, we conclude that the instrumental variables *RANK* and *POS* are valid at the 5% significance level.